

1 Introduction

In scenes containing visually similar workers, locating a helmet is only part of the task: a relational request must identify whose helmet is intended. Small equipment regions provide limited visual evidence, and darkness, dust, blur and glare can further obscure that evidence. A category-correct mask can therefore answer the wrong worker-specific question. We study this ambiguity through supplied miner identity memory: a reference identifies the worker, while a shared query requests the helmet belonging to that worker.

An aggregate segmentation score does not by itself test whether this reference controls target identity. A model can repeatedly select a plausible helmet without following a change in the requested worker. We consequently hold the observation and relational query fixed and switch only the supplied identity memory. Each switch has a corresponding helmet target. This controlled intervention distinguishes memory-dependent target selection from a preference for one visually salient object, while a localization criterion prevents identity ranking alone from being counted as successful segmentation.

Our implemented CMA design conditions segmentation on entity memory at two stages. Reference appearance and bounding-box geometry are mapped into the language-model input at the [REF] token. The associated language representation and reference-mask geometry then condition the SAM prompt used to decode the equipment mask. A paired counterfactual objective encourages a memory-conditioned helmet prediction to overlap its own target more than the other worker’s target. These mechanisms describe the implemented system; our comparison does not isolate the gain of each component.

We summarize the contributions as follows:

- We formulate counterfactual identity-memory relational perception for visually similar workers and small PPE targets, asking whether a supplied worker identity controls which equipment instance is segmented.
- We instantiate dual-stage REF conditioning through appearance/bbox input injection and reference-conditioned segmentation prompts, with a paired counterfactual identity-ranking objective. We evaluate supplied-memory use by switching identity under a fixed observation/query and measuring CMSA, Memory Fidelity, Identity Error Rate and mIoU, without attributing isolated causal gains to components that lack matched ablations.
- We provide a frozen system-level comparison under a fixed coal-mine-style compound degradation. The primary documented-protocol-disjoint confirmation shows stronger absolute degraded segmentation and identity fidelity than the pinned SegLLM comparator. Exact historical exposure remains unknown;

this is not a claim of lower degradation sensitivity or exact training-unseen generalization.

2 Method

2.1 Task and supplied identity memory

Let I be an observation, q a relational query, R_i a supplied miner reference mask, b_i its normalized bounding box, and Y_i the corresponding helmet target. We denote the supplied identity condition by $M_i = (R_i, b_i, C(I, R_i, b_i))$, where C masks the background, crops the reference box, zero-pads to a square and applies the reference-image processor. For two workers in the same image, the desired behavior is

$$f(I, q, M_A) \rightarrow Y_A, \quad f(I, q, M_B) \rightarrow Y_B. \quad (1)$$

Here memory denotes supplied identity evidence, not an autonomously maintained state. Under degradation the reference appearance is taken from the same degraded observation as the main image; the supplied mask and box identify the worker. This protocol tests memory use under degraded appearance, not recovery of an independently corrupted reference mask.

2.2 Dual-stage identity-memory conditioning

The following equations abstract the verified implementation rather than specifying every layer. First, visual tokens from the reference crop are averaged and combined with a bounding-box projection:

$$v_i = \text{mean } E_{\text{vis}}(C(I, R_i, b_i)), \quad (2)$$

$$r_i = P_{\text{in}}(v_i + B(b_i)), \quad (3)$$

$$e'_{\text{REF},i} = e_{\text{REF}} + \alpha r_i. \quad (4)$$

The frozen configuration uses $\alpha = 0.5$. This scalar addition is implemented as `gated_add`; it is not an estimated memory-reliability gate. The language model processes the image and conversation with the injected reference embedding.

For output-side conditioning, let h_i denote the projected REF-associated language hidden representation and z_i the SEG-derived text prompt. The supplied mask is bilinearly resized to 16×16 , flattened and concatenated with b_i before a geometry projection G . The memory prompt is

$$g_i = G([\text{vec}(\text{resize}_{16}(R_i)); b_i]), \quad (5)$$

$$c_i = s \text{ cap}_{20}(h_i + g_i), \quad (6)$$

$$t_i = \text{clip}_{[-50,50]}(z_i + c_i), \quad (7)$$

$$\hat{Y}_i = D_{\text{SAM}}(E_{\text{SAM}}(I), t_i). \quad (8)$$

The operator cap_{20} bounds vector norm; s is a learned reference scale, whose configured initialization is 0.2 rather than its final learned value. Validity flags restrict reference conditioning to the relevant rows. The frozen prompt

mode adds context to the text prompt before SAM prompt encoding and single-mask decoding. Its contribution is evaluated as part of the complete system, not through a component-isolated ablation.

2.3 Counterfactual identity objective

The documented training construction groups worker-helmet pairs from the same image and assigns the same second-round query to each pair. Training outputs alternate miner and helmet masks. In addition to token cross-entropy and weighted mask BCE/Dice losses, an auxiliary REF decoding branch reconstructs the supplied miner mask on valid reference rows. Writing these normalized losses as $\mathcal{L}_{\text{text}}$, $\mathcal{L}_{\text{seg}}$ and $\mathcal{L}_{\text{ref}}$, the implemented objective has the form

$$\mathcal{L} = \mathcal{L}_{\text{text}} + \mathcal{L}_{\text{seg}} + \lambda_{\text{ref}}\mathcal{L}_{\text{ref}} + \lambda_{\text{rank}}\mathcal{L}_{\text{rank}}. \quad (9)$$

For a grouped example, let p_i be the sigmoid helmet prediction conditioned on identity i . The differentiable overlap matrix and ranking term are

$$S_{ij} = \frac{\sum_x p_i(x)Y_j(x)}{\max\{\sum_x [p_i(x) + Y_j(x) - p_i(x)Y_j(x)], \epsilon\}}, \quad (10)$$

$$\mathcal{L}_{\text{rank}} = \frac{1}{N} \sum_i \left[\mu + \max_{j \neq i} S_{ij} - S_{ii} \right]_+. \quad (11)$$

Here N is the number of helmet references in the group. The code uses $\epsilon = 10^{-6}$ and selects helmet entries from the alternating mask sequence; the rank term is applied only to eligible groups with at least two helmet targets. The saved w15 configuration records $\lambda_{\text{ref}} = 1.0$, $\lambda_{\text{rank}} = 1.5$ and $\mu = 0.05$. These are hash-verified checkpoint metadata. Exact historical run-bound data and CE/BCE/Dice coefficient overrides remain unknown, so launcher defaults are not presented as verified historical settings. We make no claim that the rank term alone caused the whole-system comparison gap.

2.4 Counterfactual memory evaluation

Evaluation uses binary masks, separately from the soft training objective. Within each fixed observation/query group, define

$$J_{ij} = \text{IoU}(\hat{Y}_i, Y_j), \quad d_i = J_{ii}, \quad (12)$$

$$w_i = \max_{j \neq i} J_{ij}, \quad \Delta_i = d_i - w_i. \quad (13)$$

Memory Fidelity is the reference-wise mean of $\mathbf{1}[d_i > w_i]$; ties, including all-zero rows, fail. With $\tau = 0.5$, define $a_i = \mathbf{1}[d_i > w_i \wedge d_i \geq \tau]$. CMSA is the fraction of identity pairs for which both references succeed:

$$\text{CMSA} = \frac{1}{|\mathcal{P}|} \sum_{(i,j) \in \mathcal{P}} a_i a_j. \quad (14)$$

Pairs are formed within each group; the primary set has two identities per group, so each group contributes one

pair. Identity Error Rate is the reference-wise mean of $\mathbf{1}[w_i > d_i \wedge w_i \geq \tau]$. The mean identity margin averages Δ_i , while target mIoU averages d_i . Thus Fidelity can be correct despite poor localization, and IER is not its complement. Aggregation weights references equally for these reference-level metrics and pairs equally for CMSA.

The frozen Cycle025 runner loads base-w15 in BF16 and uses a fixed multiround conversation containing SEG placeholders. It performs a direct forward pass, extracts helmet rows, and thresholds mask logits at zero; it does not generate an autonomous dialogue or select candidates using target IoU. Helmet targets are excluded from inference raster reads and supervision-mask placeholders are zero. Predictions are frozen before offline scoring. CMA and pinned SegLLM receive the same underlying observation/query and corresponding supplied miner identity information through their method-native memory interfaces, not identical internal memory encodings. The primary protocol is disjoint from documented training/evaluation image-byte registries; exact run-bound and ancestor exposure remain unknown.

2.5 Scope boundary

The evidence evaluates supplied-memory use under a fixed compound stressor. It does not establish autonomous memory writing, persistence or repair, a successful inference-time agent controller, safety guarantees, or isolated component superiority. Earlier Layer-2 controller/verifier attempts remain negative evidence and are not part of the validated method contribution.